

6.25 Solution: (a) For a single charged particle in the atom, the Lorentz force is

$$\frac{d\mathbf{p}_i}{dt} = q_i(\mathbf{E}(\mathbf{r}_i) + \mathbf{v}_i \times \mathbf{B}(\mathbf{r}_i)),$$

where $\mathbf{p}_i$, $\mathbf{v}_i$, and $\mathbf{r}_i$ are the particle's momentum, velocity, and location. Then,

$$\frac{d\mathbf{P}_{\text{atom}}}{dt} = \sum_i \frac{d\mathbf{p}_i}{dt} = \sum_i q_i(\mathbf{E}(\mathbf{r}_i) + \mathbf{v}_i \times \mathbf{B}(\mathbf{r}_i)).$$

Expanding around the location of the atom, we have

$$\begin{aligned} \frac{d\mathbf{P}_{\text{atom}}}{dt} &= \sum_i q_i(\mathbf{E}(\mathbf{0}) + \mathbf{r}_i \cdot \nabla \mathbf{E}(\mathbf{0}) + \mathbf{v}_i \times \mathbf{B}(\mathbf{0})) \\ &= \sum_i (q_i \mathbf{r}_i) \cdot \nabla \mathbf{E}(\mathbf{0}) + \sum_i \frac{d}{dt} (q_i \mathbf{r}_i) \times \mathbf{B}(\mathbf{0}), \end{aligned}$$

where we have used the charge neutrality condition of the atom,

$$\sum_i q_i = 0.$$

Define the dipole moment of the atom as

$$\mathbf{d} = \sum_i q_i \mathbf{r}_i,$$

we will have

$$\frac{d\mathbf{P}_{\text{atom}}}{dt} = (\mathbf{d} \cdot \nabla) \mathbf{E} + \dot{\mathbf{d}} \times \mathbf{B}.$$

(b) By a uniform plane wave, electric field gradient should vanish, and we are left with

$$\frac{d\mathbf{P}_{\text{atom}}}{dt} = \dot{\mathbf{d}} \times \mathbf{B}.$$

Averaging over volume, and define the mechanical momentum and polarization as

$$\mathbf{g}_{\text{mech}} = \frac{1}{\Delta V} \int_{\Delta V} \mathbf{P}_{\text{atom}} d^3x, \quad \mathbf{P} = \frac{1}{\Delta V} \int_{\Delta V} \mathbf{d} d^3x,$$

then

$$\frac{d\mathbf{g}_{\text{mech}}}{dt} = \dot{\mathbf{P}} \times \mathbf{B}.$$

From the relation that $\mathbf{P} = (\epsilon - \epsilon_0)\mathbf{E}$, the above equation becomes

$$\frac{d\mathbf{g}_{\text{mech}}}{dt} = \mu_0 \epsilon_0 \frac{\epsilon - \epsilon_0}{\epsilon_0} \dot{\mathbf{E}} \times \mathbf{H} = \frac{1}{c^2} (n^2 - 1) \dot{\mathbf{E}} \times \mathbf{H},$$

where $n^2 = \epsilon/\epsilon_0$. Finally,

$$\dot{\mathbf{E}} \times \mathbf{H} = \frac{1}{2} \frac{d}{dt} (\mathbf{E} \times \mathbf{H}), \quad \mathbf{g}_{\text{em}} = \frac{1}{c^2} \mathbf{E} \times \mathbf{H},$$

we will have

$$\frac{d\mathbf{g}_{\text{mech}}}{dt} = \frac{1}{2} (n^2 - 1) \frac{d\mathbf{g}_{\text{em}}}{dt}.$$